T.C.
BAHÇEŞEHİR UNIVERSITY

FACULTY OF ENGINEERING AND NATURAL SCIENCES

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT FINAL REPORT

DIGITAL DICE

EEE2204 Course Project

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1 OVERVIEW

This project aims to design and implement a digital dice simulator circuit that visually displays the values of two independent dice using seven-segment displays. The system leverages basic digital electronics principles and standard integrated circuits (ICs) to simulate the random behavior of physical dice. Each die operates independently, with a separate timer-counter-decoder-display chain, ensuring random and asynchronous values similar to two physical dice rolls.

1.1 Problem Statement and Objectives

The purpose of this project is to design and build a digital dice simulator capable of visually displaying the values of two independent dice. Traditional dice rolling introduces physical randomness that can be difficult to reproduce electronically without complex systems. This project aims to provide a simple, efficient, and hardware-based solution to simulate the behavior of rolling two dice simultaneously using digital logic principles.

The main challenge lies in creating two independent random outputs, each representing a value between 1 and 6, and accurately displaying these values in a clear and reliable manner. The system must operate continuously and asynchronously, closely mimicking the behavior of two real dice rolls.

1.2 Background Information

The simulation of random number generation in electronic systems has been a longstanding challenge in the field of digital electronics. Traditional mechanical dice are widely used in gaming and educational applications due to their simplicity and the inherent randomness of each roll. However, creating an electronic equivalent requires the precise design of circuits capable of generating unpredictable yet bounded outputs.

Digital dice simulators offer a practical solution by utilizing the principles of sequential and combinational logic to replicate the random nature of dice rolls. These systems typically rely on timing signals and counting mechanisms to produce varying numerical outputs that can be visually displayed. The accurate display of these values is commonly achieved through seven-segment displays, which provide a simple and effective method for numerical representation.

This project builds upon fundamental concepts of digital circuit design, such as clock generation, counting sequences, and display decoding. It also explores the integration of these elements to achieve independent and asynchronous operation for two simultaneous dice simulations. The study of such circuits not only enhances understanding of digital logic systems but also provides hands-on experience in building functional, real-world applications without the use of microcontrollers or programmable devices.

The creation of a dual digital dice simulator exemplifies how basic digital components can be configured to solve complex design problems, making it an excellent educational tool for students and enthusiasts in the field of electronics and circuit design.

2 METHODOLOGY

2.1 Project design

The digital dual dice simulator replicates the behavior of two physical dice using basic digital logic. The system consists of two identical sub-circuits, each independently generating random values from 1 to 6 and displaying the result. A timing signal generator provides continuous clock pulses to each counting circuit, which rapidly cycles through possible dice values. This creates a random effect until the user activates a control using the push button to stop the count, effectively "rolling" the dice. The counting output is decoded and sent to visual displays, which present the values clearly. Both dice operate asynchronously, ensuring independent and realistic results. The design emphasizes modularity, reliability, and energy efficiency, showcasing the integration of sequential and combinational logic elements in a simple and educational hardware project.

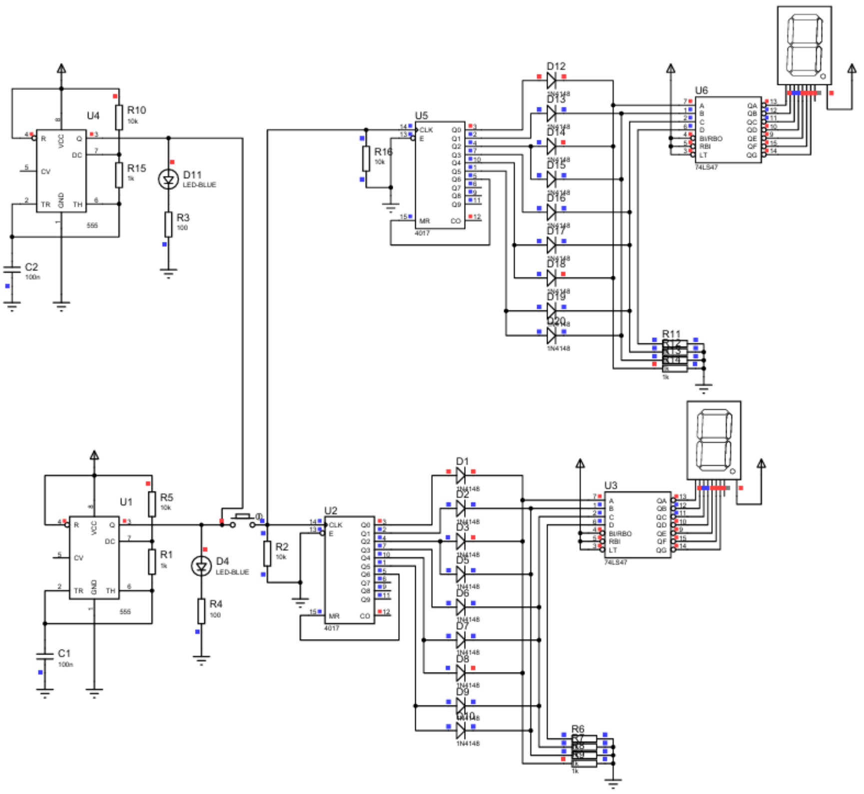


Figure 1. Overall design of the digital dice system and circuit schematic on Proteus

2.2 Project components

- **Breadboards:** Used for assembling and testing the circuit without soldering.
- **7-Segment Displays (2 pieces):** To visually display the values of each die.
- **Timer ICs (2 pieces) (555):** To generate continuous clock pulses for each die circuit.
- **Decade Counter ICs (2 pieces) (4017):** To cycle through a sequence representing dice values.
- **BCD to 7-Segment Decoder/Driver ICs (2 pieces) (7447):** To convert the counter output to display-compatible signals.
- **Push Button:** To manually control or reset the dice roll.
- **Resistors:** Used for current limiting and signal stabilization within the circuit.
- **Capacitors:** Employed for timing stability and noise reduction.
- **Jumper Wires:** For making electrical connections between components on the breadboard.
- **Power Supply (9V*2):** To provide power to the circuit.

3 WORK PLAN

3.1 Tasks and Time Line

The project was completed through equal collaboration between both of our group members (Ayse Ece, Andac).

All stages of the work, including the circuit design and assembly, component purchasing, simulation and testing, documentation, and final presentation, were carried out together as a team.

Every team member contributed actively to each phase of the project. There was no individual task division, as the group preferred to work jointly at all times to ensure accuracy and learning for all members.

	WEEKS					
TASK LIST	**1**	**2**	**3**	**4**	**5**	**6**
1. Research and Design						
1.1 Define the goals, and perform a literature search.						
1.2 Prepare an overall design and order parts.						
2. Construction and Coding						
2.1 Assemble first prototypes						
2.2 Assemble final prototypes						
2.3 Prepare Final Product						
2.4 Integrate all systems						
3. Testing and Documentation						
3.1 Prepare the proposal						
3.2 Verification						
3.3 Prepare the report						
3.4 Presentation						

Table 1. The project Gantt chart.

3.2 Cost Proposal

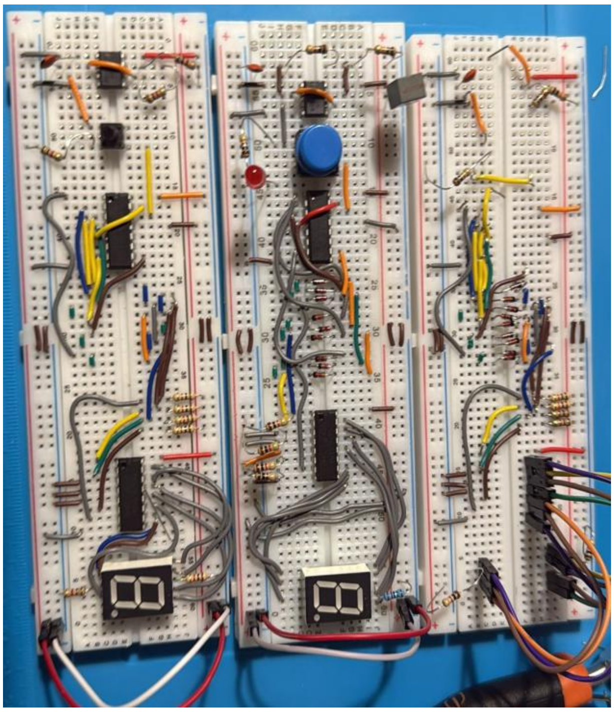
Component	Cost (TL)
ICs (7447, 4017, 555)	150
7-Segment Displays and other materials	50
Resistors, Capacitors and others	50
Breadboard, cables and other materials	200
TOTAL	**450**

Table 2. Components and their estimated costs.

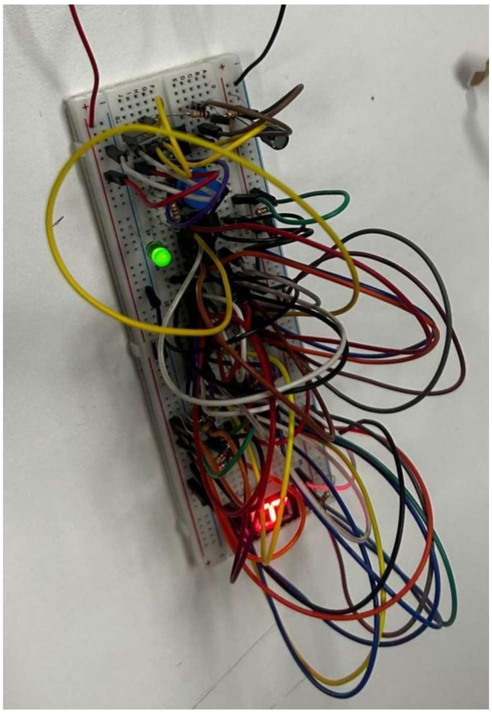
4 REFERENCES

1. Texas Instruments. *NE555 Timer Datasheet*. <https://www.ti.com/lit/ds/symlink/ne555.pdf>
2. Texas Instruments. *CD4017 Decade Counter Datasheet*. Available at: <https://www.ti.com/lit/ds/symlink/cd4017b.pdf>
3. Texas Instruments. *SN74LS47 BCD to 7-Segment Decoder Datasheet*. Available at: <https://www.ti.com/lit/ds/symlink/sn74ls47.pdf>
4. Autodesk. *Tinkercad Circuits*.: < https://www.tinkercad.com/ >
5. Labcenter Electronics. *Proteus Design Suite*. < https://www.labcenter.com/ >
6. Horowitz, P., & Hill, W. (2015). *The Art of Electronics (3rd ed.)*. Cambridge University Press.

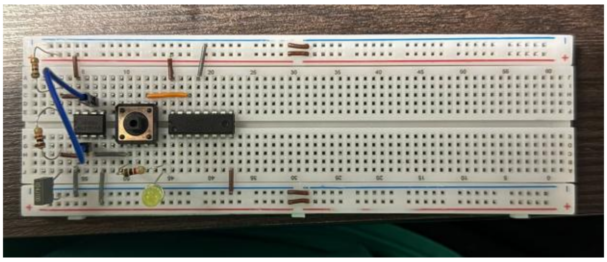
5 PHOTOS



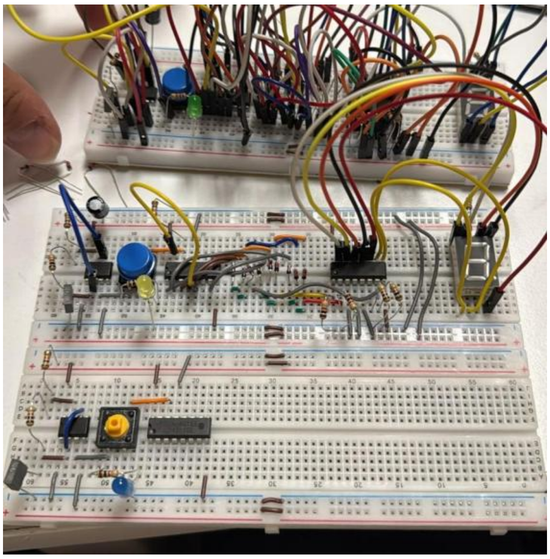
Failed Prototypes



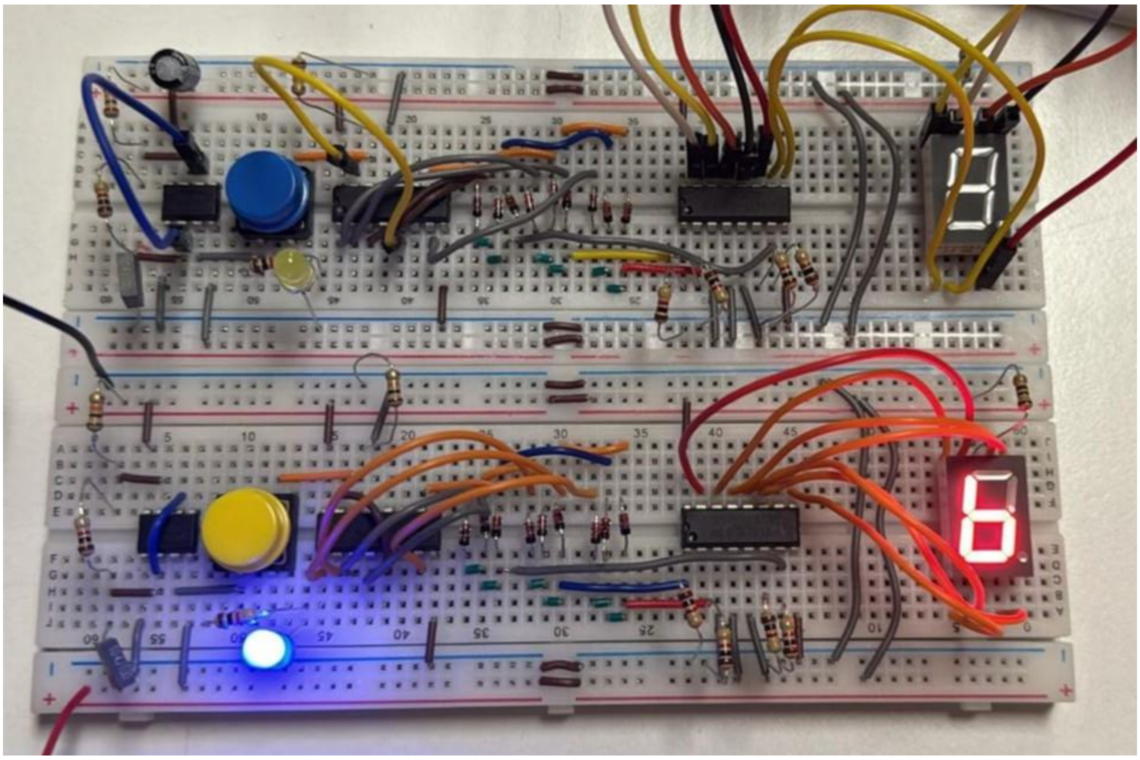
First Functional Prototype



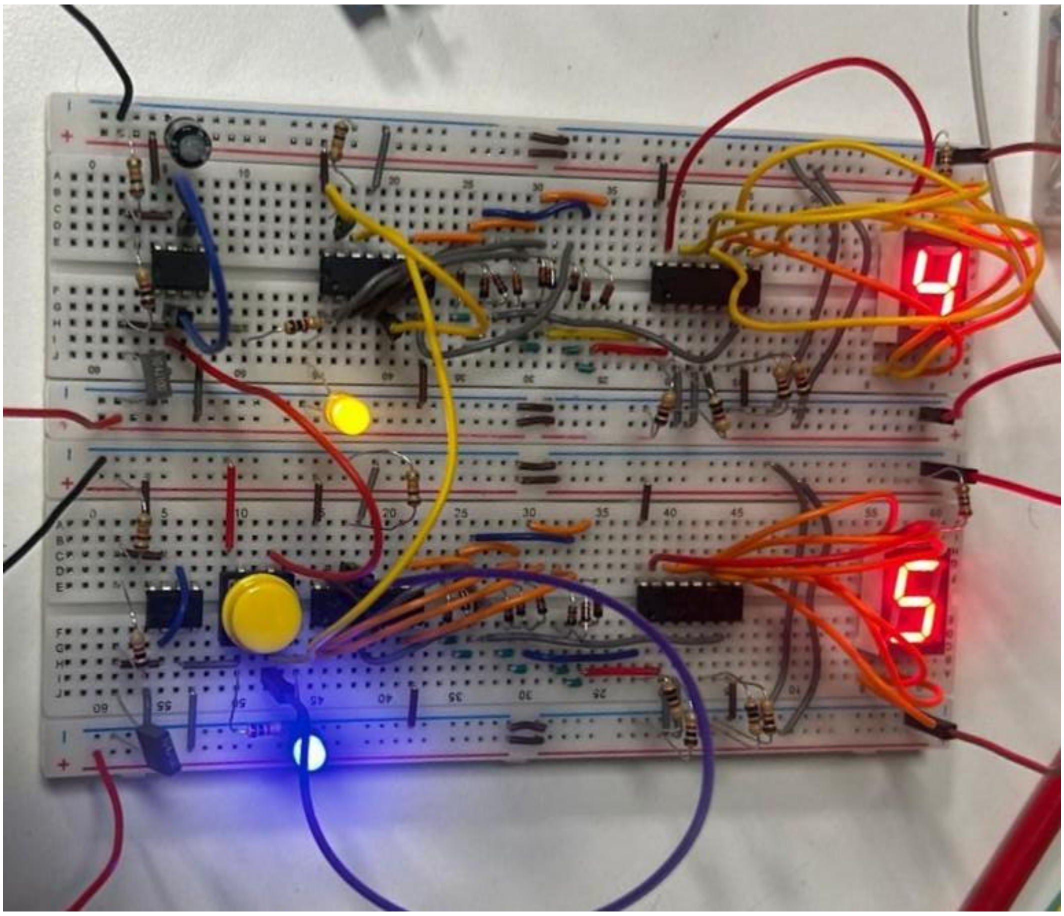
Building Second Functional Prototype



Building Third Functional Prototype

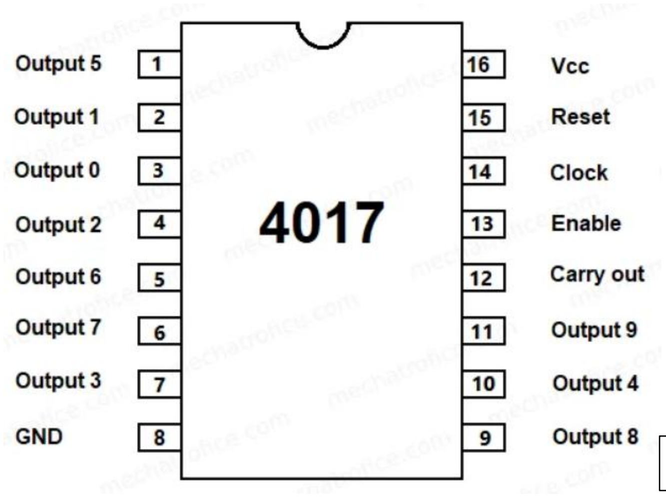


Second & Third Prototypes Before Connecting

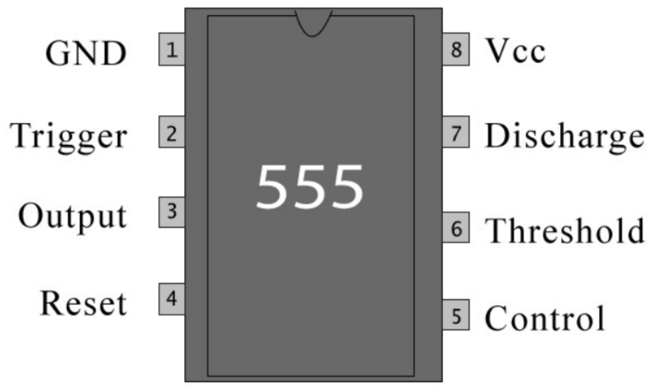


Final Product

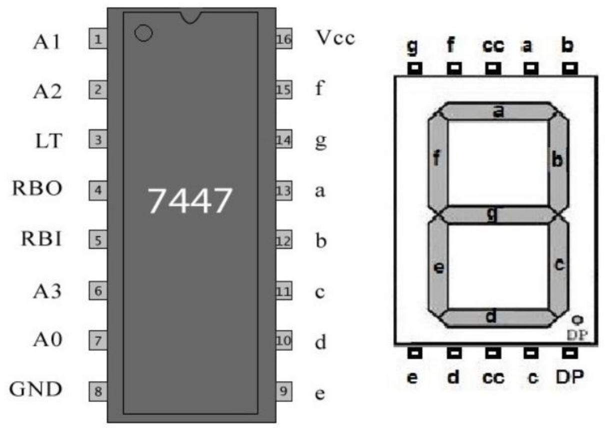
6 APPENDIX



Img. 1: IC 4017 Data SHeet



Img. 2: 555 Timer Data Sheet



Img. 3: IC 7447 & 7-Segment Display Data Sheet

555 Timer Frequency Formula

$$f = \frac{1.44}{(R_1 + 2 \times R_2) \times C_1}$$

Calculations:

$$f = \frac{1.44}{(4700 + 2 \times 10000) \times 10 \times 10^{-6}} \approx 4.8\,\text{Hz}$$